

Supplementary Material

Viewpoint Invariance Is a Theorem, Not a Fit

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This supplement carries the depth condensed out of the main paper: the proof of the invariance theorem (Sec. S1), the full invariant-set specification (Sec. S2), the eight-gate numerical certificate (Sec. S3), the complete precision budget (Sec. S4), the nondeterminism seed distribution (Sec. S5), the protocol audit and continuous-time control (Sec. S6), the units audit behind our comparison against published KIMORE numbers (Sec. S7), and the full second-corpus node-failure curve (Sec. S8). All numbers are reproduced from the same banked artifacts as the main paper; equation and proposition references (e.g. Prop. 1) point to the main text.

S1. Proof of the Invariance Theorem

Proposition 1 ($SO(3)$ -invariance of the read-out; = Prop. 1). *Let the encoder intertwine with the Wigner-D action, so under $g \in SO(3)$ each latent transforms as $s_j \mapsto s_j$ (type-0) and $v_j \mapsto D_1(g) v_j$ (type-1), with $D_1(g) \in SO(3)$ orthogonal and $\det D_1(g) = +1$. Then the projection Π (main Eq. 1–2) is invariant under $SO(3)$ for arbitrary encoder weights, and the predicted score is invariant to any rigid camera motion.*

Proof. Feeding root-relative coordinates makes every input a difference $\tilde{x}_i - \tilde{x}_j$, so a camera translation cancels; it remains to treat rotations. Consider each generator of Eq. 1 under $v \mapsto D_1(g)v$ with $R := D_1(g)$ orthogonal: (i) $\tanh s_j$ is a function of a type-0 scalar, hence unchanged. (ii) $\|v_j^c\| = \|Rv_j^c\|$ since $R^\top R = I$; so $\log(1 + \|v\|)$ is invariant. (iii) $\langle \widehat{Rv}, \widehat{Rv'} \rangle = \langle Rv/\|v\|, Rv'/\|v'\| \rangle = \langle \hat{v}, \hat{v'} \rangle$, again because R preserves inner products. (iv) $\det[\widehat{Rv^a}, \widehat{Rv^b}, \widehat{Rv^c}] = \det(R) \det[\hat{v}^a, \hat{v}^b, \hat{v}^c] = \det[\hat{v}^a, \hat{v}^b, \hat{v}^c]$, using $\det(R) = +1$; the triple products are invariant under *proper* rotations (and flip sign under a reflection, $\det = -1$, which is exactly why they narrow the guaranteed group from $O(3)$ to $SO(3)$). The bone lengths $d_{ij} = \|\tilde{x}_i - \tilde{x}_j\|$ and the anatomical signed volumes χ_τ (normalised determinants of joint triads) are invariant by the same orthogonality and

Table S1. Structural specification of the invariant cut Π .

Generator	Irrep	Count	Dim
<i>Parity-even</i> — generating set complete for $O(3)$			
scalars $\tanh s$	$0e$	n_0	32
vector norms $\log(1 + \ v\)$	$0e$	n_1	8
cosines $\langle \hat{v}, \hat{v'} \rangle$	$0e$	$\binom{n_1}{2}$	28
<i>pooled</i> mean $\oplus$ max		2×68	136
bone lengths d_{ij}	$0e$	per bone	24
even subtotal			160
<i>Parity-odd</i> — $SO(3)$ chiral completion			
triple products $\det[\hat{v}, \hat{v'}, \hat{v'']}$	$0o$	$\binom{n_1}{3}$	56
<i>pooled</i> mean $\oplus$ max		2×56	112
anatomical volumes χ_τ	$0o$	per frame	11
odd subtotal			123
full $SO(3)$ cut			283

$\det(R) = +1$ arguments. Finally mean_j and max_j are applied identically to invariant per-joint quantities, hence themselves invariant. Every entry of Π is therefore fixed by g , for any weights; the head is a deterministic function of Π , so the score is invariant. $\square$

S2. Full Invariant-Set Specification

Table S1 states both generating sets explicitly at the deployed widths ($n_0=32$, $n_1=8$; $J=25$ joints, 24 bones, $|\mathcal{T}|=11$ anatomical frames). Both parity classes share one mean $\oplus$ max joint pool; bone lengths and signed volumes are concatenated index-preserved (unpooled). The parity-even subtotal (160) is exactly the projection whose precision budget is audited in Sec. S4; the parity-odd completion adds 123 dimensions.

S3. Numerical Certificate (E1–E8)

We certify the *deployed* model. E1–E5 establish that the claimed invariance is present; E6–E8 establish that it is the one we actually have. The three conditions of E3–E4 — violation at roundoff, *flat* in angle, and $r \gg 1$ — are what

057 upgrade “small” to “zero”: magnitude alone would not. Re-
058 ported figures are from a representative fp64 run; being
059 roundoff-floor quantities they are *regime*-stable (order of
060 magnitude, sign of slope, $r \gg 10^5$) but not bit-reproducible.

061 E1 **Representation orthogonality:** $\max_g \|\rho(g)^\top \rho(g) -$
062 $I\|_F \leq 10^{-12}$ (fp64).

063 E2 **Invariance of the motion channels:** δ, σ verified invari-
064 ant, not assumed.

065 E3 **Rotation sweep:** score violation $\sim 10^{-16}$ across $\theta \in$
066 $[0.05, 3.0]$ rad, with log-log slope $|\text{slope}| < 0.5$ (flat).
067 A modelling violation grows with angle; roundoff does
068 not.

069 E4 **Precision scaling:** $r = \bar{\delta}_{\text{fp}32} / \bar{\delta}_{\text{fp}64} = 5.44 \times$
070 $10^{-8} / 4.43 \times 10^{-16} = 1.23 \times 10^8$, tracking the machine-
071 epsilon ratio; a genuine violation would be precision-
072 independent ($r \approx 1$).

073 E5 **Translation invariance:** on raw world coordinates,
074 $|s(\mathbf{x}+\mathbf{c}) - s(\mathbf{x})|/|s(\mathbf{x})| = 2.6 \times 10^{-16}$.

075 E6 **Parity:** under a reflection the parity-even cut gives *ex-*
076 *actly* 0 (bit-identical on a mirrored body — an unin-
077 tended $O(3)$ -invariance), whereas the completed cut gives
078 4.4×10^{-2} (seed 0, `certify_egru.json`). A single
079 reflection certifies the entire improper coset.

080 E7 **Laterality:** permuting left/right joint *indices* (a rela-
081 labelling, not a reflection) moves the score by 6.9×10^{-2}
082 / 5.6×10^{-2} for the two cuts (seed 0); both separate
083 a left- from a right-limb movement. E6–E7 are model-
084 dependent modelling quantities, not roundoff-floor ones,
085 and vary across seeds.

086 E8 **The fix is free:** read-out invariance under Haar-random
087 *proper* rotations (up to $\theta=\pi$), with the parity-odd chan-
088 nels active, is 2.9×10^{-16} — unchanged. E1–E5 are like-
089 wise over Haar-random $SO(3)$, so the certified invariance
090 is to the full rotation group, of which a camera azimuth or
091 the NTU cross-view displacement is one instance. We re-
092 port no separate NTU-specific invariance measurement:
093 the gates are architectural and the corpus does not change
094 them.

095 S4. Complete Precision Budget

096 The full fp64→int8 budget — including the bf16 and int8-
097 activations-only rows — is reported in the main paper’s de-
098 ployment table. The int8 cliff is attributable *solely* to ac-
099 tivation grid-rounding: weight-only quantization preserves
100 the theorem (a quantized model is a different equivariant
101 model), and fp16 is $3.2\times$ tighter than bf16 because equiv-
102 ariance pays in mantissa bits, which bf16 trades for expo-
103 nent range.

Table S2. Per-(seed,fold) clean MAD, $SO(3)$ EGRU. No retraining beyond the three seeds.

fold	seed 0	seed 1	seed 2	std	gap
0	7.120	6.778	6.737	0.210	0.383
1	6.680	6.124	5.804	0.443	0.876
2	6.976	7.345	6.623	0.361	0.722
3	6.762	7.363	6.093	0.635	1.270
4	6.469	7.714	6.415	0.735	1.299
mean across-seed std				0.48	

S5. Nondeterminism: Seed Distribution

The 0.33 MAD floor quoted in the main paper is the tighter *fixed-configuration* run-to-run component (init held constant, atomic nondeterminism only). Table S2 gives the wider *seed-to-seed* distribution ($SO(3)$ EGRU clean MAD per fold across three initializations): mean across-seed std 0.48, worst pairwise gap 1.30. Both exceed every model-vs-model clean gap in the main accuracy table, so those gaps are ties. cuDNN-deterministic modes remove the cuDNN-GRU atomic component but not the initialization variance; the structural degradation columns are weight-independent (Prop. 1 holds for arbitrary weights), so the flat viewpoint curve stayed exactly flat across every seed while absolute MAD moved.

S6. Protocol Audit and Continuous-Time Control

Protocol audit. The single-exercise audit — selection bias $+1.2 \pm 0.4$ MAD (18.6% over three seeds), paired subject-cluster bootstrap $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$, clearing zero in only one of three seeds) — is reported per-seed in the main paper. It is *not* subject leakage (each subject contributes one recording on that slice, so a random split is already subject-disjoint); the slice simply carries no robust subject-level signal, so we pool the five exercises and gate every experiment on a per-exercise mean-predictor floor.

Continuous-time control. We implement the Neural CDE variant and certify it to the same E1–E8 standard; it satisfies an exact equivariance certificate and *fails* its own measured floor (8.43 vs. 8.25 MAD), the cleanest demonstration that a structural guarantee is worthless without measured signal. Analysing the adaptive solver, a Dormand–Prince controller accepts/rejects steps on a per-component scaled-RMS norm that is *not* group-invariant even when the vector field is exactly equivariant, so rotated and unrotated integrations receive different step grids and de-align; a per-irrep-normalised norm removes this, driving grid divergence to identically zero.

Irregular-sampling null (derived). A model-free spec-

$\widehat{ y } = 100 \text{ MAD/MAPE}$	Ex1	Ex2	Ex3	Ex4	Ex5
Du et al. 2021	39.4	36.6	32.8	34.1	33.2
Deb et al. 2022	41.5	60.8	51.4	42.1	39.0
Yao et al. 2023	40.2	35.1	32.5	35.5	36.1
Mourchid et al. 2023	39.5	77.3	34.3	38.1	32.8
Zhang et al. 2025	41.2	51.6	38.4	42.2	35.0
Kuang et al. 2025	43.2	31.4	41.0	30.6	32.2
Kuang et al. 2026	54.5	40.1	48.3	63.4	18.8
Point-cloud transf.	34.1	29.6	38.1	33.4	32.4
true score mean	40.7	35.6	39.5	35.0	36.4
true mean $ z $	0.80	0.87	0.79	0.86	0.84

Table S3. Target magnitude recovered from each paper’s own published MAD/MAPE pair, against KIMORE’s true clinical Total Score statistics. Every row lands in the score band (30–63), none anywhere near the standardised band (≈ 0.8). Rows reporting no MAPE are omitted.

tral census finds 97.7% of KIMORE’s *positional* energy below the fixed-grid resampling corner (though 70% of *velocity* energy lies above it), so resampling preserves pose geometry while low-passing the derivative band and acts as a denoiser on healthy-form exercises, where the discriminative signal is positional — the null was predictable before the experiment, though the boundary would move for tremor-dominated tasks. The mechanism is nonetheless real: a physiologically calibrated 5 Hz tremor is recovered from the true irregular stamps at $r=1.000$ under drop rates up to 70%, while a resampled estimator decays to 0.785.

S7. Units of the Published KIMORE Numbers

The main text compares our MAD against published KIMORE MAD directly. That is only legitimate if both are in clinical-score units, and none of those papers states the scale. The question is not academic: the reference point-cloud transformer standardises its targets with a `StandardScaler` ($\sigma = 8.466$ on `Kimore_ex1`) and its per-epoch training print is in standardised units while its evaluator inverse-transforms, so the two readings of the same headline number differ by $8.5\times$.

The published tables settle it without recourse to anyone’s source code. Every method in this lineage reports MAD and MAPE for the same run, with $\text{MAPE} = 100 \cdot \text{mean}(|y - \hat{y}|/|y|)$, so

$$\widehat{|y|} = 100 \cdot \text{MAD/MAPE} \quad (1)$$

recovers the magnitude of whatever array was handed to the metric. On raw clinical scores it must land near KIMORE’s per-exercise means; on standardised targets it must land near $\text{mean}|z| \approx 0.8$.

Table S3 is unambiguous: the median recovered magnitude over all 40 published cells is 38.1 against a true per-

exercise mean of 35.0–40.7 (median relative error 10.6%), while the standardised reading is excluded by at least $70\times$ — it would require these papers to have printed MAPE values in the tens of percent rather than the 0.17–6.0 they report. The published KIMORE MAD numbers are therefore in 0–50 clinical-score units, and our 6.3–6.7 is on the same axis.

The convention is also documented at its source, so the inference above is corroborated rather than load-bearing. Both papers load KIMORE through descendants of a single `DataLoader`, the one released with the STGCN-rehab benchmark (Deb et al., IEEE TNSRE 2022); the point-cloud transformer’s copy is near-identical line for line. That release’s training script inverse-transforms predictions *and* targets before computing MAD, RMSE and MAPE, and its released label file holds raw clinical Total Scores. The reporting convention is therefore fixed upstream of any individual paper, and Tab. S3 confirms it independently, from published numbers alone. The audit is reproduced by `src/published_units_audit.py`, which gates on all three conditions.

S8. Node-Failure Curve and Which Spread We Quote

The main paper summarises second-corpus node failure with one number, AUROC lost from $k=0$ to $k=8$ frozen joints. That scalar hides the shape, and the shape is not uniformly in our favour.

REHAB24-6, AUROC at k frozen joints	$k=0$	$k=1$	$k=2$	$k=4$	$k=8$
EGRU (ours)	0.738	0.714	0.694	0.659	0.606
PCT (baseline)	0.707	0.697	0.685	0.660	0.606

Table S4. The full curve behind the single “AUROC lost” figure. Mean over 3 seeds $\times$ 5 subject-disjoint folds.

Table S4 shows the two models *cross* between $k=2$ and $k=4$: we start 0.031 ahead, are level at $k=4$ (0.659 vs. 0.660), and are 0.021 behind at $k=8$. This is the same Pareto picture the main paper states for KIMORE — the point-cloud transformer is the more node-robust model, and our claim is a trade, not a dominance. Reporting only the endpoint difference would have made the trade look like a loss on one axis rather than a crossover with a locatable knee.

Which spread the $\pm$ denotes. These runs admit two variances that differ by roughly $5\times$, and they answer different questions. The *fold-level* sd — the spread of 5 subject-disjoint folds within one seed — is 0.060 (EGRU) and 0.065 (PCT); a fold holds out 2 of 10 subjects, so this is large by construction. The *seed-level* sd — the spread of the 3 per-seed means — is 0.010 and 0.023, because averaging five

218 folds cancels most of the subject variation. The $\pm$ quoted in
219 the main paper is the seed-level one: it describes *seed repro-*
220 *ducibility*, not confidence in the AUROC given this subject
221 pool. At 10 subjects the fold-level figure is the honest guide
222 to how far these numbers would move on a different co-
223 hort, which is why this corpus is presented as a structural
224 replication and not as an accuracy benchmark. Both are re-

225
generated by `src/second.corpus.tables.py`.